MILITARY SPECIFICATION

TRANSCEIVER: 10540501

This amendment forms a part of Military Specification MIL-T-60500A(MU), dated 22 September 1970.

Paragraph 3.5.4: Delete and substitute the following:

"3.5.4 Humidity.- The transceiver shall be capable of withstanding 10 cycles of exposure to the following:

a. Two hours of increasing temperature to +105°F and 95 percent relative humidity.

b. Sixteen hours at +150°F and 90 percent relative humidity.

c. Two hours decreasing temperature from +150°F to +70°F and increasing relative humidity to 95 percent.

d. Four hours at +70°F and 95 percent relative humidity.

3.5.4.1 Subsequent to exposure to humidity, the transceiver shall not have been damaged or adversely affected and shall meet the requirements of 3.5.9 through 3.7.10.2 within 30 minutes after exposure."

Paragraph 3.6 - Sealing: In the second sentence after the word "entering" add "(gray colored port)" and after the word "bled" add "(yellow colored port)".

Paragraph 3.7: Delete lines "a" and "b" in the second sentence and substitute the following:

"a.	21 to 30 volts dc	5 amps max.
b.	7 to 10 volts dc	1 amp max."

Paragraph 3.7.1a, lines 1 and 2: Delete "nominal 24 Vdc" and substitute "21 Vdc".

Paragraph 3.7.1b, line 1: Delete "nominal 9 VDC" and substitute "7 Vdc".

Paragraph 3.7.3.1b: Delete in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

Paragraph 3.7.5.6 Low voltage indicator: Delete sentence and substitute the following: "With the power switch in the "ON" position and the READ switch depressed, the numeral 2 shall appear in the unit column in addition to Numbers 0, 1, 5 or 7 at a voltage between 20 and 22 volts inclusive and shall remain on for voltages less than 20 volts down to the voltage below which the nixie tubes fail to operate."

* Paragraph 3.7.8: Delete and substitute the following:

"3.7.8 Electromagnetic Compatibility.- The radiated and conducted electromagnetic emissions from the transceiver shall not exceed the allowable limits of tests CE03 and RE02 of MIL-STD-461 for class IIA equipment. Switch transients and short duration interference shall be exempted from this requirement."

Paragraph 3.7.10.2a: Delete in its entirety.

Paragraph 4.3.2.1: Delete all after second sentence and add new paragraph as follows:

"4.3.2.2 Transceiver.- Examination and test related to Section 3 herein shall be performed in accordance with Tables I through V. Examination and tests for packaging, packing and markings shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated characteristics and sampling in Tables I to V inclusive shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot."

Table I - Classification of Defects: Delete Characteristic 135. At bottom of table add: "NOTE" Unless otherwise specified, tests shall be conducted between 70° to 80°F.

Paragraph 4.3.4 Certification: Delete in its entirety the characteristic and test data table at bottom of paragraph.

Paragraph 4.4.1.2, at the end of the first sentence, add "in accordance with 4.6.5.8.1."

Paragraph 4.4.2, Table IV: Delete and substitute the following:

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Storage temperature (-80°F and +160°F)	3.5.3	4.6.2.1
307.	Operational temperature (-65°F and +155°F)	3.5.9	4.6.2.9
308.	Vibration	3.5.1	4.6.2.3

Source: https://assist.dla.mil - Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

NOTE: Unless otherwise specified, tests shall be conducted between 70° to 80°F.

Add Paragraph 4.4.2.1 as follows: "4.4.2.1.- Three (3) transceivers shall be selected at random as a special sampling at 25%, 50% and 75% completion of contract production. Each sample shall be subjected to the tests in Table V and shall then meet the requirements and tests in Table I.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
309.	Rain	3.5.8	4.6.2.8
310.	Shock (bet/70° to 80°F)	3.5.2	4.6.2.2
311.	Humidity	3.5.4	4.6.2.4
312.	Sand and dust	3.5.5	4.6.2.5
313.	Salt spray	3.5.6	4.6.2.6"

Paragraph 4.6.2.2: Delete second sentence and substitute, "The test shall be performed as specified in Procedure I of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.2.

Paragraph 4.6.2.3: Delete fourth sentence and substitute, "The vibration test shall be performed in accordance with Figure 514-1, Curve C of MIL-STD-810, except double amplitude shall not exceed .036 inch."

Paragraph 4.6.2.4: Delete and substitute the following:

"4.6.2.4 Humidity.- The humidity test shall be performed as outlined in MIL-STD-810, Procedure V to determine compliance with the conditions specified in 3.5.4."

Paragraph 4.6.2.5: Add, "During the sand and dust test, all exposed optical elements shall be protected with a masked covering.

Paragraph 4.6.5, lines 4 and 5: Delete "a nominal 24.0 ± 2.0 volts d.c." and "9.0 ± 0.5 volts d.c." and substitute "21.0 ± 0.5 volts d.c." and "7.0 ± 0.5 volts d.c." respectively.

Paragraph 4.6.5.2.1: Delete the last two sentences.

Paragraph 4.6.5.2.2: Add the following sentence to the beginning of the paragraph: "Close parallax shield over receiver optics."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

Paragraph 4.6.5.8.1, lines 1 and 2: Delete and substitute "Three (3) items shall be selected as follows from the transceivers to be presented as the first article sample (see 4.2) to determine compliance with 3.7.9 and subsequent inspection in accordance with Table III."

Paragraph 4.6.5.8.1a, last line: Delete "and final assembly of 3 transceivers" and substitute of "Of 3 transceivers as outlined below, subsequent to which, the contractor shall assemble the selected groups into 3 complete transceivers for inspection and testing in conformance with 4.6.5.8.2."

Paragraph 4.6.5.8.2: Delete the first 3 sentences and substitute: "The 3 transceivers obtained from 4.6.5.8.1 shall be subjected to all requirements and tests of this specification in compliance with 3.7.9. Subsequent to selection (4.6.5.8.1), the conduct of final assembly, examination and tests shall be witnessed by the quality assurance representative. Interchangeability tests (including first article) shall be conducted at each extreme voltage condition specified in 3.7.2 and b, i.e.:

21 + 0.5 Vdc (5 amps. max.) and 7.0 + 0.5 Vdc (1 amp. max.) - low extremes

30 - 0.5 Vdc (5 amps. max.) and 10 - 0.5 Vdc (1 amp. max.) - high extremes

After Government approval and acceptance of the first article (4.2), interchangeability tests shall be conducted in accordance with sampling in 4.4.1.2 and Table III."

Figure 3: Delete and substitute attached Figure 3.

* The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

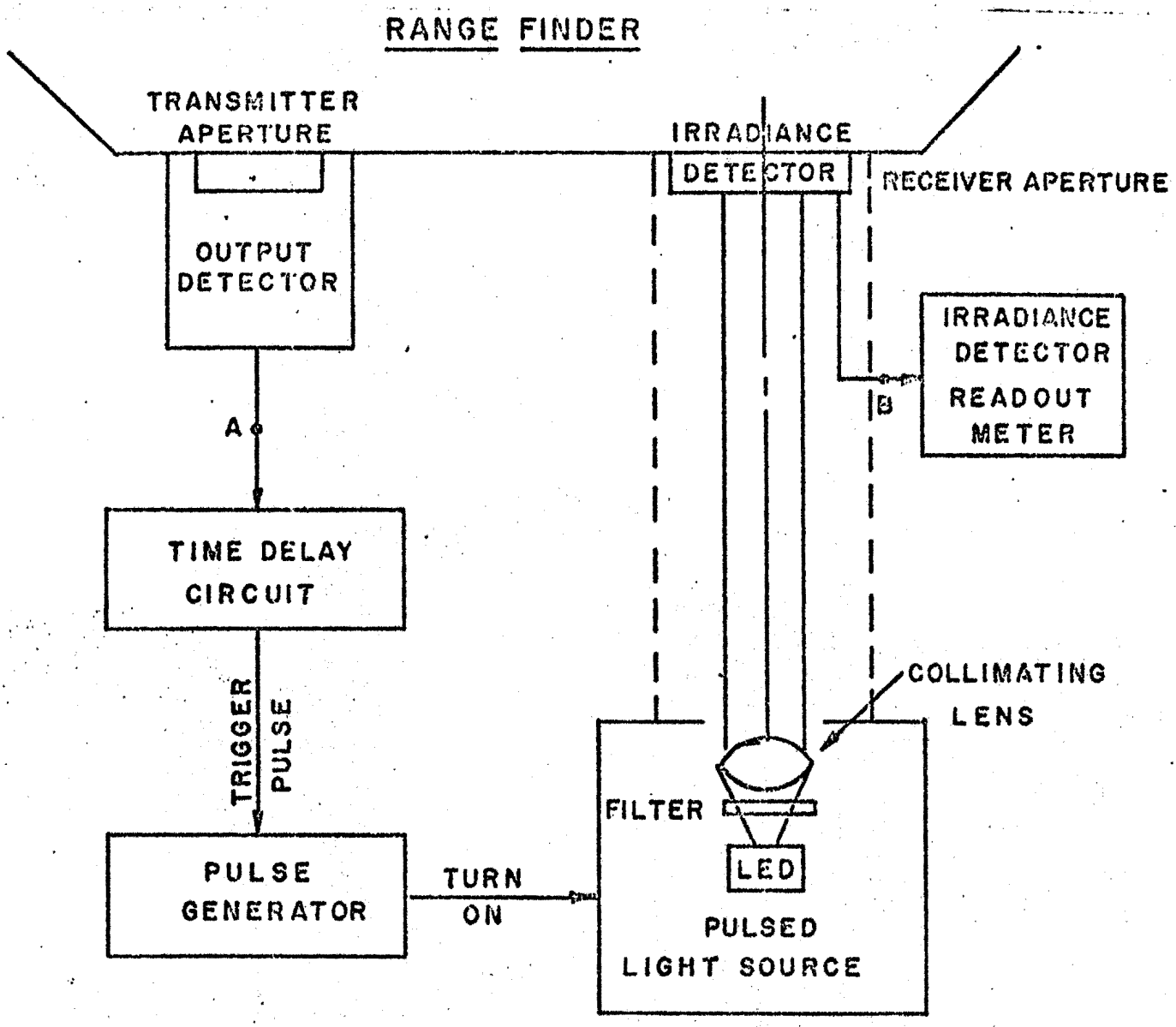
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A518

Source: https://assist.dla.mil - Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.



NOTE: 1. MEASURE TIME DELAY FROM POINT "A" FROM OUTPUT DETECTOR TO POINT "B" FROM IRRADIANCE DETECTOR.

2. ISOLATE THE RECEIVER APERTURE TO PREVENT ERRONEOUS READINGS FROM REFLECTIONS FROM OUTPUT PULSE.

RECEIVER SENSITIVITY TEST SETUP

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.